

Yuexin (Oscar) Luo

MASTER'S STUDENT, MECHANICAL ENGINEERING (CONTROL & ROBOTICS), CARNEGIE MELLON UNIVERSITY

Pittsburgh, PA, USA

+1 540-750-5654 | oluo@andrew.cmu.edu | <https://oscar-prise.github.io/> | [yuexin-luo-496149250](https://www.linkedin.com/in/yuexin-luo-496149250)



LinkedIn



Portfolio

Education

M.S. Carnegie Mellon University Mechanical Engineering – Control & Robotics | GPA 3.96/4.0 | Expected May 2027
B.S. Virginia Tech Mechanical Engineering – Robotics & Mechatronics | GPA 3.36/4.0 | May 2023

Research Experience

Metamobility Lab, Carnegie Mellon University

Pittsburgh, PA, USA

HIP EXOSKELETON LEAD DESIGNER

Sep. 2025 – Present

- Converted commercial hip exoskeleton to accept custom motors/controllers and implemented delayed-output feedback control, reducing metabolic cost by 1%
- Reverse-engineered the commercial hip exoskeleton, reducing device weight from 4.2kg to 3kg within 3 months
- Diagnosed hip-joint motor rocking under high torque to excessive thigh-interface clearance via 26 build-test-analyze iterations in one semester
- Designed and manufactured calf-surrounding thigh structures, raising torque capacity before motor rocking from 12 N-m to 17 N-m
- Cut required belts and straps by half through the redesigned thigh structure

Work Experience

Soochow University Embedded System Lab

Suzhou, Jiangsu, China

LAB ASSISTANT

Apr. 2024 – Mar. 2025

- Developed synchronous serial communication between two MCUs and a host computer using SPI as part of a graduate research project
- Developed a C# UI to translate data between characters, binary, decimal, and hex, improving debugging efficiency
- Prepared 68 PCBs by soldering electrical components for an undergraduate course

Hounen Solar America Inc.

Orangeburg, SC, USA

EQUIPMENT ENGINEER

Jul. 2023 – Dec. 2023

- Maintained and operated a solar cell stringer soldering machine, reducing unplanned downtime by 40%
- Analyzed and reduced assembly line blockage from twice a day to once per three days using FMEA
- Crafted standardized operating procedures for 4 machines, cutting employee training time by an average of 50%
- Instructed over 12 employees on machine operation along the assembly line

Robotic Projects

DJI RoboMaster Competition, Virginia Tech

Blacksburg, VA, USA

MECHANICAL ENGINEER, "HERO" SUB-TEAM

Sep. 2021 – May 2023

- Redesigned golf-ball launcher feed path from bottom-fed to rear-fed, cutting weight by 67% and assembly time by 40% while maintaining projectile capacity
- Built a mobile welded test chassis to field-test the launcher's target lock-on robustness under repeated handling
- Instructed 4 new team members on converting 3D CAD models into 2D CNC-manufacturable parts

Senior Capstone Design, MELD Automation Prep Line

Christiansburg, VA, USA

MECHANICAL ENGINEER

Aug. 2022 – May 2023

- Designed a pneumatic trap door to release cut bar stock into the next subsystem on cut detection, completing the transfer within 13% of the allotted cycle time
- Iterated corner-bracket geometry through several FEA-guided design cycles to meet the factor-of-safety requirement with a 2% margin
- Fabricated motor brackets by CNC mill and MIG welder as part of a cutting subsystem that can hold up to 3x desired mass

Intro Human-Robot Interaction Course Project, Best Pokémon Trainer

Blacksburg, VA, USA

ENGINEER

Mar. 2023 – May 2023

- Extracted Pokémon statistics to establish a database in Python, widening AI-generated Pokémon variation by around 400%
- Recovered reward weights with a standard deviation of 0.025 based on the Metropolis-Hastings method in Python
- Used inverse reinforcement learning (IRL) to train an AI with a 72% win rate against a simple AI and 45% against a human player

Skills

Software	SolidWorks Fusion 360 Siemens NX MATLAB
Programming Languages	Python C/C++ C# Linux CLI PowerShell
Manufacturing	Rapid Prototyping 3-D Printing Soldering Mill & Lathe Machining Welding
Embedded Systems	STM32 Jetson Arduino